

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 2.3: Plant Gaseous Exchange and Respiration — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Plant Gaseous Exchange and Respiration
Driving Question	DRIVING QUESTION: How do plants "breathe," and what happens inside their cells when they release the energy stored in food?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Predict Phase: Where Do the Bubbles Come From?	Students observed freshly cut sukuma wiki leaves submerged in warm (~42°C) and cold (~15°C) water, recording the location and rate of bubble emergence on annotated leaf diagrams. They noted that bubbles appeared to emerge from specific point-sources on the leaf surface — more abundantly from warm water than cold — rather than uniformly across the entire leaf.	Students learned that gases are present inside plant leaves and can be released when the leaf is submerged in water; that the rate of gas release is affected by water temperature; and that bubble production is concentrated at specific sites on the leaf rather than the whole surface. Students also learned that scientists use models, annotated diagrams, and public prediction-tracking to organise their thinking.	Students constructed initial explanations (revised from their pre-observation predictions) proposing that the bubbles come from openings or spaces inside the leaf, and that warmth increases the rate at which gas escapes. Students could not yet fully explain what the gas is, why only specific spots produce bubbles, or how temperature mechanistically increases bubble production — these gaps are recorded on the DQB as productive open questions for the rest of the unit.
Lesson 2 Sites of Gaseous Exchange in Plants	Bubbles escaped from specific spots on the submerged sukuma wiki leaf surface, not uniformly from the whole leaf. The animation showed gas molecules moving in and out through distinct openings on the leaf and stem surfaces. Fewer bubbles appeared in cold water compared to warm water.	Plants exchange gases through four main sites: stomata (pores on the epidermis controlled by guard cells), the waxy cuticle (limited exchange), lenticels (on woody stems and branches, e.g., mango bark), and pneumatophores (specialised roots in waterlogged areas like Mida Creek mangroves). Stomata are the primary site of gas exchange in leaves and are responsible for the bubble locations observed in Lesson 1.	The bubbles in Lesson 1 appeared at specific spots because those spots are stomata — microscopic pores surrounded by guard cells that can open and close. Stomata are concentrated on the lower epidermis of most leaves. The reduction in bubbles in cold water suggests that low temperature affects the opening of stomata or the rate of gas release, a question the class will investigate further in upcoming lessons.
Lesson 3 Stomatal Mechanisms:	In the animations, we observed that guard cells change shape when ions and water	We learned that two theories explain stomatal movement: the potassium ion	We can now explain why bubbles in Lesson 1 came from specific spots — those spots

How Guard Cells Control Gas Exchange	move in or out of them. In Lesson 1, we observed that warm water produced more bubbles from specific spots on sukuma wiki leaves.	theory (K^+ is actively pumped into guard cells, lowering water potential, causing osmotic water entry and turgidity, opening the pore) and the starch-sugar hypothesis (starch converts to soluble sugars in light, similarly driving water entry by osmosis). Both theories agree that osmosis and turgor pressure drive guard cell shape changes.	are stomata whose guard cells were opened by ion and water movement. We can also partially explain why warm water produced more bubbles: higher temperatures may increase the rate of ion pumping and osmotic movement, leading to more open stomata and greater gas release. The wilting maize near Kisumu closes stomata to conserve water by reversing these ion movements, then reopens at cooler dusk.
Lesson 4 Aerobic & Anaerobic Respiration — Research, Experiment & Data Analysis	Students observed that limewater connected to the aerobic yeast tube (Tube A, open to air) turned milky-white/turbid faster and more intensely than the limewater connected to the anaerobic yeast tube (Tube B, sealed). They also observed that warming increased the rate of turbidity change, mirroring the warm-water effect on sukuma wiki bubble production from the anchoring phenomenon.	Students learned that all living cells — including plant cells and yeast — release energy from glucose through respiration. Aerobic respiration ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$) occurs in the mitochondria and produces more energy and more CO_2 per glucose molecule. Anaerobic respiration ($C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{energy}$) occurs in the cytoplasm without oxygen and produces less energy. Both pathways produce CO_2 , which must exit the plant through the gas-exchange structures (stomata, lenticels) studied in Lessons 1–3.	The tiny bubbles streaming from the sukuma wiki leaves submerged in warm water were partly CO_2 produced by cellular respiration in the leaf cells (in addition to oxygen from photosynthesis). Warm water increases enzyme activity in the leaf cells, which speeds up the rate of respiration, producing CO_2 more quickly — hence more bubbles in warm water than in cold water. The CO_2 produced inside leaf cells diffuses out through the same stomata that regulate gaseous exchange, connecting the cellular (mitochondrion) level to the tissue (stomata) level of organisation.
Lesson 5 Importance of Anaerobic Respiration: From Busaa to Biogas	Students observed: (1) six Kenyan case studies (uji, busaa, mandazi, biogas, ethanol, waterlogged roots) during the gallery walk, identifying substrate, agent, products, and oxygen conditions at each station; (2) balloon inflation on yeast-sugar-water bottles, providing real-time evidence of CO_2 gas production in the absence of oxygen; (3) comparison of balloon inflation rates across different yeast concentrations (Bottles A, B, C).	Students learned: (1) anaerobic respiration occurs in the absence of oxygen, producing ethanol and CO_2 in yeast and plant roots (alcoholic fermentation) or lactic acid in bacteria and animal muscle; (2) anaerobic respiration yields only 2 ATP per glucose molecule compared to 36–38 ATP in aerobic respiration; (3) fermentation by yeast underpins economically important Kenyan processes including uji souring, busaa brewing, mandazi leavening, biogas production, and industrial ethanol manufacture; (4) plants shift to anaerobic respiration in their roots when soil is waterlogged and oxygen is depleted — connecting plant cell biochemistry to agricultural and ecological contexts.	Students explained: (1) the mandazi dough phenomenon — CO_2 produced by yeast during anaerobic fermentation of sugar creates gas pockets that cause the dough to rise, exactly as CO_2 produced during aerobic respiration in sukuma wiki leaves caused the bubbles observed in the anchoring phenomenon; (2) the word equation for anaerobic respiration in yeast: Glucose $\rightarrow$ Ethanol + Carbon dioxide (+ 2 ATP); (3) why the same glucose molecule can follow two different pathways depending on oxygen availability, and why the anaerobic pathway is less energetically efficient but essential under oxygen-limited conditions.
Lesson 6 Factors Affecting Respiration Rate:	Yeast in warm sugar-cane juice produced significantly more CO_2 bubbles per minute than yeast in cold water; increasing glucose	Respiration rate in yeast — and by extension in plant cells — is controlled by temperature (through its effect on enzyme	The anchoring phenomenon of more bubbles from sukuma wiki in warm water than in cold water can now be explained

Temperature, Glucose, and Oxygen	concentration increased bubble rate up to a point; sealed anaerobic bottles produced rapid early bubble bursts but the rate declined quickly compared to open aerobic setups.	kinetic energy and collision frequency), glucose concentration (through substrate availability for respiratory enzymes), and oxygen availability (which determines whether the aerobic or anaerobic pathway is used and therefore how much ATP is produced).	quantitatively: warmer temperatures increase the kinetic energy of respiratory enzyme molecules and their substrates, increasing the frequency of successful enzyme-substrate collisions in the glycolysis and Krebs cycle pathways, thereby increasing the rate of CO ₂ production and the number of bubbles observed — directly linking cellular enzyme kinetics to the visible macroscopic phenomenon first observed in Lesson 1.
Lesson 7 Adaptations for Gaseous Exchange in Different Environments	Learners observed prepared slides and hand specimens of three Kenyan plant types — an acacia/cactus xerophyte (Rift Valley/Turkana), water hyacinth hydrophyte (Lake Victoria, Kisumu), and sukuma wiki mesophyte (Nairobi school garden) — and recorded structural features including stomata position, cuticle thickness, aerenchyma presence, and specialised root structures in a three-column comparison table.	Learners learned that gaseous-exchange adaptations vary systematically with environment: xerophytes have sunken stomata and thick cuticles to minimise water loss in arid conditions; hydrophytes have stomata on the upper (air-facing) leaf surface and internal aerenchyma to channel gases to submerged tissues; mesophytes like sukuma wiki have a moderate arrangement suited to Kenya's temperate highland conditions — and that all three solutions address the same fundamental challenge of moving gases in and out efficiently while managing water loss.	Learners used the Claim–Evidence–Reasoning framework to explain how each structural adaptation is functionally suited to its ecosystem, revised their cumulative plant gaseous-exchange model by adding three new ecosystem panels (Turkana xerophyte, Lake Victoria hydrophyte, Nairobi mesophyte), and updated the Driving Question Board with evidence-based claims and new questions — advancing their collective answer to 'How do plants exchange gases with their environment?' and setting up the transition to cellular respiration in Lessons 8–12.
Lesson 8 Photosynthesis, Gaseous Exchange & Respiration Linkages: Systems Synthesis and Model Building	Learners observed their own Evidence Card Sets summarising data from Lessons 1–7, sorted cards into process categories and an overlap zone, and reviewed peer Systems Diagrams during the Gallery Walk — all anchored to the original sukuma wiki bubble phenomenon from Mwalimu Njoroge's class.	Learners synthesised that photosynthesis, gaseous exchange, and cellular respiration are not isolated processes but an interdependent system: CO ₂ and O ₂ cycle between photosynthesis and respiration; gaseous exchange structures (stomata, lenticels, cuticle) regulate molecular entry and exit; glucose produced in photosynthesis fuels respiration; and the rate of the system — evidenced by bubble production in warm water — is influenced by temperature acting on enzyme-driven reactions.	Learners explained the sukuma wiki anchoring phenomenon at a systems level: bubbles stream from stomata because O ₂ produced in photosynthesis exits through these structures; warm water increases reaction rates in enzyme-driven photosynthesis and respiration pathways, producing more O ₂ and CO ₂ and thus more bubbles; and the entire gas exchange apparatus of the leaf — guard cells, mesophyll air spaces, chloroplasts, and mitochondria — functions as a coordinated system to sustain the plant's energy and gas balance.
Lesson 9 Project Launch & Planning: Investigating Anaerobic Respiration with Local Kenyan Materials	Students observed three set-up stations representing anaerobic respiration in local Kenyan contexts: (A) a yeast and sugar solution with a balloon inflating over the bottle neck, (B) a biogas simulation bottle with vegetable scraps (sukuma wiki offcuts,	Students learned that anaerobic respiration is the process by which cells release energy from glucose in the absence of oxygen; that in yeast cells this produces ethanol and carbon dioxide (as seen in rising mandazi dough and balloon inflation), while in	Students explained that the gas bubbles seen in the sukuma wiki phenomenon (Lesson 1) are produced by aerobic respiration in plant cells — a process requiring oxygen. In contrast, the gas produced in yeast fermentation (mandazi

	ugali remnants) producing gas displacement, and (C) two jars of uji at different fermentation stages with pH strips and hand lenses. Students also observed the anchoring phenomenon image (sukuma wiki bubbles) placed alongside a mandazi dough image to prompt comparison.	bacteria and some plant cells it may produce lactic acid; that anaerobic respiration produces less ATP than aerobic respiration; that locally available Kenyan materials (unga wa chachu, sukari, sukuma wiki, uji base) can be used to investigate fermentation; and that a valid scientific investigation requires a testable hypothesis, clearly identified variables, and a structured data collection plan.	dough rising, balloon inflation) comes from anaerobic respiration — the same starting molecule (glucose) is broken down, but without oxygen, producing ethanol and CO₂ instead of water and CO₂. Students also explained the design of their group investigations by formulating hypotheses in 'If-Then-Because' format and identifying variables, connecting their experimental design to the broader unit question of how living cells extract energy from organic food molecules.
Lesson 10 Project Execution: Carrying Out Investigations on Plant Gaseous Exchange and Respiration	Students observed the outcomes of their own investigations — for example: bubble production rates from sukuma wiki leaf discs at different temperatures; stomatal counts on upper vs. lower leaf surfaces using nail varnish peels; colour changes in hydrogencarbonate indicator or lime water when exposed to germinating githeri seeds; or temperature differences between germinating and boiled seeds. They recorded raw data in tables, noted anomalous readings, and documented the behaviour of leaf material, seeds, and indicator solutions over time.	Students learned how to execute a fair-test investigation by actively controlling variables and recording data with appropriate units and precision. They applied unit knowledge — stomata, guard cells, lenticels, cellular respiration, enzyme activity, gas diffusion — to explain patterns in their own data. They learned that real scientific data is sometimes messy or unexpected, and that anomalies are evidence to be explained, not discarded. They also learned to structure their thinking using the CER framework, distinguishing a claim from evidence and from reasoning.	Students explained how the biological structures and processes they studied in Lessons 1–9 (stomatal regulation by guard cells, gas exchange through lenticels and stomata, cellular respiration releasing CO₂ and energy from glucose) directly account for the patterns in their data. They connected their group findings to the anchoring phenomenon — specifically explaining why the sukuma wiki leaves in Mwalimu Njoroge's class produced bubbles from specific points (stomata), why warm water increased bubble production (higher temperature increases respiration and enzyme activity rates), and why blocking the lower surface had a greater effect than blocking the upper surface (more stomata on the abaxial surface).
Lesson 11 Project Presentations and Peer Review: Communicating Evidence on Plant Gaseous Exchange and Respiration	Groups presented their investigation findings on anaerobic respiration using yeast fermentation, biogas simulation, and food fermentation with locally available Kenyan materials. Peers observed that CO₂ production was confirmed across multiple independent investigations, and that conditions such as temperature, glucose availability, and oxygen exclusion affected gas production rates in consistent ways.	Scientific confidence increases when multiple independent investigations converge on the same finding. Plants and yeast release CO₂ during cellular respiration through both aerobic and anaerobic pathways. This CO₂ exits plant cells through stomata, lenticels, and other gaseous exchange sites — the same sites from which bubbles emerged from the sukuma wiki leaves in Lesson 1. Argumentation requires citing specific evidence, not just asserting claims.	The bubbles observed streaming from specific points on the sukuma wiki leaves in warm water were CO₂ produced by cellular respiration in mesophyll cells, exiting through stomata and lenticels. Warm water increased metabolic and diffusion rates, producing more bubbles. Collective investigation evidence confirms that plants do exchange gases with their environment continuously — through respiration at all times, and through photosynthesis when light is available — making the answer to the driving question a synthesis of all unit learning.
Lesson 12	Tiny bubbles streaming from specific spots	The bubbles emerge specifically from	The sukuma wiki leaves were not 'breathing'

Consolidation, Model Finalisation & Unit Review: How Do Plants "Breathe" and Release Energy?	on sukuma wiki leaves submerged in warm water — not from the entire leaf surface, and far fewer bubbles appearing when the water was cold.	stomata — microscopic pores on the leaf surface controlled by guard cells. The gases in the bubbles are CO₂ (produced by cellular respiration in mesophyll cells) and O₂ (produced by photosynthesis in chloroplasts). Warm water increases bubble production because higher temperature increases enzyme activity, accelerating both aerobic respiration (producing more CO₂) and, where light is available, photosynthesis (producing more O₂). Cold water slows enzyme-catalysed reactions, reducing gas production. Plants exchange gases not only through stomata but also through the cuticle, lenticels on woody stems, and pneumatophores in waterlogged environments. Inside mesophyll cells, aerobic respiration proceeds through glycolysis (cytoplasm), the Krebs cycle (mitochondrial matrix), and oxidative phosphorylation (inner mitochondrial membrane), yielding approximately 36–38 ATP molecules per glucose. When oxygen is absent, anaerobic respiration produces ethanol and CO₂ (or lactate), yielding only 2 ATP per glucose. Photosynthesis and respiration are complementary processes: the glucose and O₂ produced by photosynthesis serve as substrates for respiration, and the CO₂ and H₂O produced by respiration are raw materials for photosynthesis.	randomly — they were performing precisely regulated, enzyme-driven metabolic processes at the cellular level. The specific spots from which bubbles emerged are stomata: openings whose size is actively regulated by guard cells through changes in turgor pressure driven by potassium ion (K⁺) transport. The bubbles themselves are the visible, macroscopic signature of microscopic chemistry: aerobic cellular respiration in the mitochondria of mesophyll cells releasing CO₂, and photosynthesis in chloroplasts releasing O₂. The temperature of the water acted as a throttle on enzyme activity — warm water accelerated the enzymes of both respiration (e.g., pyruvate dehydrogenase, enzymes of the Krebs cycle) and photosynthesis (e.g., RuBisCO), producing more gas, and therefore more bubbles. This observation — puzzling in Lesson 1, fully explained by Lesson 12 — demonstrates that plants do 'breathe' in a deep biochemical sense: they continuously exchange gases with their environment and convert stored chemical energy into usable ATP through the same fundamental respiration chemistry that powers all living cells, from sukuma wiki leaves in a Nairobi school garden to the cells of a Rift Valley marathon runner.
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ATTRIBUTION AND LICENSING

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